Introduction to Analysis II

Lecture notes, Chapters 6–10

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Chapter 6. Sequences and Series of Functions

6.1 Uniform convergence

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Theorem 6.4 (6.2.1)

Let $f_n : [a,b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a,b]$;
(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t| \, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$ this shows that (f_n) is uniformly Cauchy, hence converges uniformly to some f. Fix x and set $\phi_n(t) = \frac{f_n(t)-f_n(x)}{t-x}$, $\phi(t) = \frac{f(t)-f(x)}{t-x}$ for $t \neq x$. By $(*)$, (ϕ_n) is uniformly Cauchy on $[a,b] - \{x\}$, so $\phi_n \to \phi$ uniformly there. Let $g = \lim f_n'$. Given ϵ, choose N with $|\phi_N - \phi| < \epsilon$ on $[a,b] - \{x\}$ and $|f_N'(x) - g(x)| < \epsilon$, and δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$ for such t, so $f'(x) = g(x)$. ∎

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a,b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b-a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b-a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b-a)$. ∎

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n\ge 0}(-1)^n t^{2n}$ (uniform on $[-r,r]$, $r < 1$) gives $\arctan x = \sum (-1)^n x^{2n+1}/(2n+1)$ for $|x| < 1$; similarly $\log(1+x) = \sum (-1)^{n+1} x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $na_n \to 0$: $\frac{n}{2} a_n \le a_{\lceil n/2 \rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn} - c_n| < \epsilon$ for $m \ge N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. Regard $f_m(n) = a_{mn}$ as functions on $X = \{1, 1/2, 1/3, \dots\} \cup \{0\}$ (with $n \leftrightarrow 1/n$), extended to 0 by $f_m(0) = b_m$; each f_m is continuous by (a), and by (c) and Theorem 6.2 the f_m converge uniformly on X (at 0 too, since $|b_m - b_k| \le \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1), with value c_n at $1/n$ and $\lim b_m$ at 0; continuity at 0 gives $\lim_n c_n = \lim_m b_m$. ∎

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \le \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \le \sum_m b_m$, so the inner sums converge absolutely. Let $S_{MN} = \sum_{m \le M} \sum_{n \le N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m \le M} \sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n \le N} \sum_m a_{mn}$, uniformly in N, since $|S_{MN} - \sum_{n \le N} \sum_m a_{mn}| \le \sum_{m > M} b_m$. Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \ge 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m \le M} \sum_{n \le N} a_{mn} \le \sum_{n \le N} \sum_m a_{mn} \le \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\le$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n) \in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k \in \mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n \to \infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \ge 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k \in \mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ■

6.4 Power series

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup |a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup |a_n x^n|^{1/n} = \lambda |x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup |n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived

series has the same radius and converges uniformly on $[-r,r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ■

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R,R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!}(x-c)^n$ for $|x-c| < R-|c|$.

Proof. $f(x) = \sum_n a_n(c+(x-c))^n = \sum_n \sum_{k\le n} a_n \binom{n}{k} c^{n-k}(x-c)^k$. Since $\sum_n \sum_k |a_n| \binom{n}{k} |c|^{n-k}|x-c|^k = \sum_n |a_n|(|c|+|x-c|)^n < \infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x) = \sum_k (\sum_n \binom{n}{k} a_n c^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ■

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n = \sum_{k=0}^n a_k b_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$, then $\sum c_n = AB$.

Proof. With partial sums A_n, B_n, C_n and $\beta_n = B_n - B$, $C_n = \sum_{k\le n} a_k B_{n-k} = A_n B + \gamma_n$, $\gamma_n = \sum_{k\le n} a_k \beta_{n-k}$. Let $\alpha = \sum |a_k|$ and $\epsilon > 0$; choose N with $|\beta_m| < \epsilon$ for $m \ge N$. Then $|\gamma_n| \le \epsilon\alpha + \sum_{n-N<k\le n} |a_k||\beta_{n-k}|$, and the last sum tends to 0 as $n \to \infty$ because $a_k \to 0$ and the β's are bounded. So $\gamma_n \to 0$ and $C_n \to AB$. ■

Theorem 6.15 (6.4.4, Abel)

Let $f(x) = \sum a_n x^n$ on $(-R,R)$. If $\sum a_n R^n$ converges, then $\lim_{x\to R^-} f(x) = \sum a_n R^n$.

Proof. Replacing a_n by $a_n R^n$ we may take $R = 1$; subtracting a constant, assume $s = \sum a_n = 0$. With partial sums $s_n \to 0$, summation by parts gives $\sum_{n\le N} a_n x^n = (1-x)\sum_{n<N} s_n x^n + s_N x^N$, so $f(x) = (1-x)\sum s_n x^n$ for $|x| < 1$. Given ϵ, choose M with $|s_n| < \epsilon$ for $n \ge M$; then $|f(x)| \le (1-x)\sum_{n<M} |s_n| + \epsilon$ for $0 < x < 1$, which is $< 2\epsilon$ for x near 1. ■

Corollary 6.16 (6.4.5)

If $\sum a_n = A$, $\sum b_n = B$ and their Cauchy product $\sum c_n = C$ all converge, then $C = AB$.

Proof. For $|x| < 1$ the series $\sum a_n x^n$, $\sum b_n x^n$ converge absolutely, so $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$ by Proposition 6.14. Let $x \to 1^-$ and use Theorem 6.15 three times. ■

Example 6.4

Abel's theorem gives $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$ and $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ from Example 6.3. Defining $e^x = \sum x^n/n!$, Proposition 6.14 gives $e^{x+y} = e^x e^y$; with $\sin, \cos$ defined by their series, $(\sin)' = \cos$, $(\cos)' = -\sin$, and $\sin^2 + \cos^2 = 1$ follow by termwise differentiation.

6.5 Trigonometric series

A complex vector space V with a map $\langle \cdot, \cdot \rangle$ that is linear in the first variable, satisfies $\langle y, x\rangle = \overline{\langle x, y\rangle}$, and $\langle x, x\rangle > 0$ for $x \neq 0$, is an **inner product space**; $\|x\| = \langle x, x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x,y\rangle| \le \|x\|\|y\|$ in a complex inner product space; hence $\|x+y\| \le \|x\| + \|y\|$.

Proof. If $y \neq 0$, put $\lambda = \langle x,y\rangle/\|y\|^2$; then $0 \le \|x-\lambda y\|^2 = \|x\|^2 - |\langle x,y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x,y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k\le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{†}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By (†), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ∎

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ∎

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx}-1}{e^{ix}-1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{‡}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi-\delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum |a_n - a_{n-1}| < \infty$, then for every $\delta \in (0,\pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi-\delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by (‡)) and constant $g_n = a_n$; take real and imaginary parts. ∎

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) Given ϵ, uniform convergence gives $|\sum_{n/2\le k\le n} a_k \sin kx| < \epsilon$ for all x and all large n. Take $x = \pi/(2n)$: for $n/2 \le k \le n$, $kx \in [\pi/4, \pi/2]$, so $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$. There are at least $n/2$ such k, so $\frac{n}{2}a_n/\sqrt{2} < \epsilon$. Hence $na_n \to 0$.

($\Leftarrow$) Since a_k decreases, (†) with $f_k = \sin kx$ and (‡) give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m/\sin(x/2) \le 2\pi a_m/x$ for $0 < x \le \pi$ (using $\sin(x/2) \ge x/\pi$). Let $\epsilon_m = \sup_{k\ge m} ka_k \to 0$. By oddness and periodicity it suffices to bound $R_m(x) = \sum_{k\ge m} a_k \sin kx$ for $x \in (0,\pi]$; let N be the largest integer

$\le \pi/x$. If $m > N$, then $m \ge N+1 > \pi/x$, so $|R_m(x)| \le 2\pi a_m/x < 2ma_m \le 2\epsilon_m$. If $m \le N$, split $R_m = \sum_{k=m}^{N} + \sum_{k>N}$. In the first sum $|a_k \sin kx| \le k a_k x \le \epsilon_m x$, and there are at most $N \le \pi/x$ terms, so it is at most $\pi\epsilon_m$; the second is at most $2\pi a_{N+1}/x < 2(N+1)a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi+2)\epsilon_m \to 0$. ■

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \le g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \le M$. Then $\sum f_n g_n$ converges uniformly.

Proof. Apply (†) to the tails: with $R_k = \sum_{j=m}^{k} f_j$ ($R_{m-1} = 0$), $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k(g_k - g_{k+1})$. Given ϵ, for m large $|R_k(x)| < \epsilon$ for all $k \ge m$, x, by (a). Since $g_k - g_{k+1} \ge 0$, the sum is at most $\epsilon M + \epsilon(g_m - g_n) \le 3M\epsilon$ in absolute value. ■

Chapter 7. Function Spaces

7.1 The space $C(X)$

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|\lambda v\| = |\lambda|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$; V is then a **normed space**, with distance $\|u - v\|$, and convergence, Cauchy sequences, open and closed sets are defined as in $\mathbb{R}^n$. A normed space in which every Cauchy sequence converges is **complete** (a **Banach space**). For compact $X \subset \mathbb{R}^n$, $C(X)$ is the space of continuous $f : X \to \mathbb{C}$ with $\|f\|_{\sup} = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1 (7.1.1)

For compact X, every Cauchy sequence in $C(X)$ converges: $C(X)$ is complete.

Proof. A Cauchy sequence in $C(X)$ is uniformly Cauchy, so converges uniformly (Theorem 6.2) to a function that is continuous (Theorem 6.1). ■

The Heine–Borel theorem fails in $C(X)$: the functions $f_n(x) = x^n$ on $[0, 1]$ form a closed bounded set (no subsequence converges uniformly, since the pointwise limit is discontinuous). The missing ingredient is equicontinuity. A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$ and $\|x - y\| < \delta$; it is **bounded** if $\sup_{f\in\mathcal{F}} \|f\|_{\sup} < \infty$.

Proposition 7.2 (7.1.2)

If X is compact and (f_n) converges uniformly in $C(X)$, then $\{f_n\}$ is equicontinuous.

Proof. Given ϵ, choose N with $\|f_n - f_N\|_{\sup} < \epsilon/3$ for $n \geq N$ (Theorem 6.2). The finitely many functions $f_1, \dots, f_N$ are uniformly continuous (Theorem 3.18 of Volume I), so some δ works for all of them with $\epsilon/3$. For $n > N$ and $\|x - y\| < \delta$, $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon$. ■

Theorem 7.3 (7.1.3, Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let (f_n) be in $\mathcal{F}$, $|f_n| \leq M$. X has a countable dense subset $E = \{p_1, p_2, \dots\}$ (for each k, finitely many balls of radius $1/k$ cover X; take their centres in X). The bounded sequence $(f_n(p_1))$ has a convergent subsequence $(f_{1,n})$; $(f_{1,n}(p_2))$ has a convergent subsequence $(f_{2,n})$; and so on. The diagonal sequence $g_n = f_{n,n}$ is, from the k-th term on, a subsequence of $(f_{k,n})$, so $(g_n(p_k))$ converges for every k. Given ϵ, take δ from equicontinuity for $\epsilon/3$, and finitely many $p_{k_1}, \dots, p_{k_r}$ with every $x \in X$ within δ of one of them. Choose N with $|g_m(p_{k_j}) - g_n(p_{k_j})| < \epsilon/3$ for $m, n \geq N$ and all j. For any x, pick $p = p_{k_j}$ with $\|x - p\| < \delta$; then $|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(p)| + |g_m(p) - g_n(p)| + |g_n(p) - g_n(x)| < \epsilon$. So (g_n) is uniformly Cauchy and converges in $C(X)$ (Theorem 7.1). ■

Combined with Proposition 7.2: a subset of $C(X)$ has compact closure iff it is bounded and equicontinuous.

Theorem 7.4 (7.1.4, Weierstrass approximation)

For every continuous $f : [a, b] \to \mathbb{R}$ there are polynomials $P_n \to f$ uniformly on $[a, b]$.

Proof. By an affine change of variable take $[a, b] = [0, 1]$, and subtracting the linear function $f(0) + x(f(1) - f(0))$ assume $f(0) = f(1) = 0$. Extend f by 0 outside $[0, 1]$; it is uniformly continuous on $\mathbb{R}$. Let $Q_n(x) = c_n(1 - x^2)^n$ on $[-1, 1]$ with c_n chosen so that $\int_{-1}^{1} Q_n = 1$. Since $(1 - x^2)^n \geq 1 - nx^2$,

$$\int_{-1}^{1} (1 - x^2)^n dx \geq 2 \int_{0}^{1/\sqrt{n}} (1 - nx^2) dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}},$$

so $c_n < \sqrt{n}$, and for $\delta \leq |x| \leq 1$, $Q_n(x) \leq \sqrt{n}(1 - \delta^2)^n \to 0$ uniformly. Let

$$P_n(x) = \int_{-1}^{1} f(x + t) Q_n(t)\, dt = \int_{0}^{1} f(t) Q_n(t - x)\, dt \qquad (0 \leq x \leq 1),$$

a polynomial in x. Given ϵ, choose δ with $|f(y) - f(x)| < \epsilon/2$ for $|y - x| < \delta$, and let $M = \max|f|$. Then

$$|P_n(x) - f(x)| \leq \int_{-1}^{1} |f(x + t) - f(x)| Q_n(t)\, dt \leq 2M \cdot 2\sqrt{n}(1 - \delta^2)^n + \frac{\epsilon}{2} < \epsilon$$

for large n, uniformly in $x \in [0, 1]$. ∎

For complex-valued f, approximate real and imaginary parts. Consequently $C[a, b]$ has a countable dense set (polynomials with rational coefficients).

7.2 Sequence spaces

$\ell^1 = \ell^1(\mathbb{N})$ is the space of sequences $a = (a(n))$ with $\|a\|_1 = \sum|a(n)| < \infty$, and ℓ^2 those with $\|a\|_2 = (\sum|a(n)|^2)^{1/2} < \infty$. ℓ^2 is a vector space with inner product $\langle a, b\rangle = \sum a(n)\overline{b(n)}$, which converges absolutely since $|a(n)b(n)| \leq \frac{1}{2}(|a(n)|^2 + |b(n)|^2)$; $\|\cdot\|_2$ is the associated norm (Theorem 6.17).

Theorem 7.5 (7.2.1)

ℓ^1 is complete.

Proof. Let (a_m) be Cauchy. Since $|a_m(n) - a_k(n)| \leq \|a_m - a_k\|_1$, each $(a_m(n))_m$ converges to some $a(n)$. Given ϵ, choose N with $\|a_m - a_k\|_1 < \epsilon$ for $m, k \geq N$. For each L, $\sum_{n \leq L}|a_m(n) - a_k(n)| < \epsilon$; let $k \to \infty$ and then $L \to \infty$: $\sum_n |a_m(n) - a(n)| \leq \epsilon$ for $m \geq N$. So $a = a_m - (a_m - a) \in \ell^1$ and $a_m \to a$. ∎

Theorem 7.6 (7.2.2)

ℓ^2 is complete.

Proof. As in Theorem 7.5: the coordinates converge, and $\sum_{n \leq L}|a_m(n) - a_k(n)|^2 < \epsilon^2$ passes to the limit in k and then in L. ∎

The same proof shows that ℓ^∞ (bounded sequences with the sup norm) is complete. The unit vectors e_n in ℓ^2 form a bounded sequence with $\|e_m - e_n\|_2 = \sqrt{2}$, so without convergent subsequence: closed bounded sets in infinite-dimensional spaces need not be compact.

7.3 Improper integrals

Let I be an interval and $f : I \to \mathbb{C}$ Riemann integrable on every compact subinterval. If $I = [a, b)$ ($b \le \infty$), the **improper integral** $\int_I f = \lim_{c\to b^-} \int_a^c f$ when the limit exists; similarly at a left end point, and for open intervals both ends are treated separately. We write $\int_I |f| < \infty$ when this improper integral converges (**absolute convergence**).

Proposition 7.7 (7.3.1)

If $0 \le f \le g$ on I and $\int_I g$ converges, then $\int_I f$ converges.

Proof. $c \mapsto \int_a^c f$ is increasing and bounded by $\int_I g$, so it has a limit as $c \to b^-$ (as for monotone sequences). ■

Proposition 7.8 (7.3.2, comparison test)

If $|f| \le g$ and $\int_I g$ converges, then $\int_I |f|$ and $\int_I f$ converge and $|\int_I f| \le \int_I |f| \le \int_I g$.

Proof. $\int_I |f|$ converges by Proposition 7.7. Cauchy criterion: for $a < c < c' < b$, $|\int_c^{c'} f| \le \int_c^{c'} |f| \to 0$ as $c, c' \to b$, so $\lim \int_a^c f$ exists (as in Theorem 2.14 of Volume I). The inequality passes to the limit from finite intervals (Theorem 5.5 of Volume I). ■

Example 7.1

$\int_1^\infty x^{-p} dx < \infty$ iff $p > 1$, and $\int_0^1 x^{-p} dx < \infty$ iff $p < 1$. $\int_0^\infty \frac{\sin x}{x}\, dx$ converges (integrate by parts: $\int_1^c \frac{\sin x}{x} dx = \frac{\cos 1}{1} - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2} dx$) but not absolutely (on $[k\pi, (k+1)\pi]$, $\int |\sin x|/x \ge \frac{2}{(k+1)\pi}$).

Proposition 7.9 (7.3.3, integral test)

If $f : [1, \infty) \to (0, \infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=2}^N f(n) \le \int_1^N f \le \sum_{n=1}^{N-1} f(n)$, and if $f \to 0$, then $\sum_{n\le N} f(n) - \int_1^N f$ converges.

Proof. On $[n, n+1]$, $f(n+1) \le f \le f(n)$; integrate and sum. The differences $d_N = \sum_{n\le N} f(n) - \int_1^N f$ decrease ($d_N - d_{N+1} = \int_N^{N+1} f - f(N+1) \ge 0$) and are $\ge f(N) > 0$, so converge. ■

For $f(x) = 1/x$ this gives Euler's constant $\gamma = \lim(1 + \frac{1}{2} + \cdots + \frac{1}{N} - \log N)$.

Theorem 7.10 (7.3.4)

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_x^y |f| < \epsilon$ whenever $x < y$ in I and $y - x < \delta$.

Proof. Choose a compact $[c, d] \subset I$ with $\int_{I-[c,d]} |f| < \epsilon/2$; f is bounded on $[c, d]$, say by K. With $\delta = \epsilon/(2K + 1)$, $\int_x^y |f| \le \int_{[x,y]\cap[c,d]} |f| + \int_{I-[c,d]} |f| < K\delta + \epsilon/2 < \epsilon$. ■

7.4 Spaces of integrable functions

Proposition 7.11 (7.4.1)

For a normed space X the following are equivalent:
(a) every Cauchy sequence in X converges;
(b) whenever $\sum\|x_n\| < \infty$, the series $\sum x_n$ converges in X.

Proof. (a)⇒(b): the partial sums are Cauchy, since $\|\sum_{m<k\le n} x_k\| \le \sum_{m<k\le n}\|x_k\|$.

(b)⇒(a): given a Cauchy (y_n), choose $n_1 < n_2 < \cdots$ with $\|y_m - y_n\| < 2^{-k}$ for $m, n \ge n_k$. With $x_1 = y_{n_1}$, $x_{k+1} = y_{n_{k+1}} - y_{n_k}$, $\sum\|x_k\| < \infty$, so $y_{n_k} = \sum_{j\le k} x_j$ converges; a Cauchy sequence with a convergent subsequence converges (Lemma 2.13 of Volume I, same proof). ∎

Let $\mathcal{R}^1(I)$ and $\mathcal{R}^2(I)$ be the locally Riemann integrable $f : I \to \mathbb{C}$ with $\int_I|f| < \infty$, resp. $\int_I|f|^2 < \infty$, with $\|f\|_1 = \int_I|f|$ and $\|f\|_2 = (\int_I|f|^2)^{1/2}$. These are seminorms ($\|f\| = 0$ does not force $f = 0$), and $\mathcal{R}^2$ carries the inner product $\langle f, g\rangle = \int_I f\bar{g}$. Neither space is complete; this is the reason for introducing the Lebesgue integral (Chapter 10). A **step function** is a finite linear combination of characteristic functions of bounded intervals.

Lemma 7.12 (7.4.2)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_2 < \epsilon$.

Proof. Treating real and imaginary parts separately, let f be real. Choose a compact $[a, b] \subset I$ with $\int_{I-[a,b]}|f|^2 < \epsilon^2/4$. On $[a, b]$, f is Riemann integrable, so bounded by some K; choose a partition with $U(f, P) - L(f, P) < \epsilon^2/(8K)$ and let $s = m_i$ on the i-th interval (and 0 off $[a, b]$). Then $0 \le f - s \le 2K$ on $[a, b]$, so $\int_a^b|f - s|^2 \le 2K\int_a^b(f - s) \le 2K(U - L) < \epsilon^2/4$. Altogether $\|f - s\|_2^2 < \epsilon^2/2$. ∎

Lemma 7.13 (7.4.3)

For a step function s on I and $\epsilon > 0$ there is a continuous g with $\|s - g\|_2 < \epsilon$ (and $g = 0$ outside a compact subinterval).

Proof. It suffices to treat $s = \chi_{[a,b]}$ (with any type of end points). Let g be 1 on $[a + \eta, b - \eta]$, 0 outside (a, b), and linear in between; then $|s - g| \le 1$ and differs from 0 only on two intervals of length η, so $\|s - g\|_2^2 \le 2\eta$. ∎

Theorem 7.14 (7.4.4)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a continuous g with $\|f - g\|_2 < \epsilon$.

Proof. Combine Lemmas 7.12 and 7.13 with the triangle inequality for $\|\cdot\|_2$. ∎

Corollary 7.15 (7.4.5)

For $f \in \mathcal{R}^2[a, b]$ there are polynomials P_n with $\|f - P_n\|_2 \to 0$.

Proof. Choose continuous g with $\|f - g\|_2 < \epsilon$ and a polynomial P with $\|g - P\|_{\sup} < \epsilon$ (Theorem 7.4); then $\|g - P\|_2 \le \epsilon\sqrt{b - a}$. ∎

Theorem 7.16 (7.4.6)

For $f \in \mathcal{R}^1(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_1 < \epsilon$, and a continuous g vanishing outside a compact subinterval with $\|f - g\|_1 < \epsilon$. For $f \in \mathcal{R}^1[a, b]$ there are polynomials P with $\|f - P\|_1 < \epsilon$.

Proof. As in Lemma 7.12, with $\int_a^b |f - s| = \int_a^b (f - s) \leq U - L$ for the lower step function. For the continuous approximation of $\chi_{[a,b]}$ the error in $\|\cdot\|_1$ is at most η. Polynomials: $\|g - P\|_1 \leq (b - a)\|g - P\|_{\sup}$. ■

Chapter 8. Functions Defined by Integrals

8.1 Integrals depending on a parameter

For f on $[a,b]\times[c,d]$ let $\phi(x)=\int_c^d f(x,t)\,dt$. Write $D_1 f=\partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f:[a,b]\times[c,d]\to\mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. (a) f is uniformly continuous on the compact rectangle (Theorem 3.18 of Volume I): given ϵ there is δ with $|f(x,t)-f(x',t)|<\epsilon$ for $|x-x'|<\delta$ and all t; then $|\phi(x)-\phi(x')|\le\epsilon(d-c)$.

(b) Fix x and a compact $[x-\eta,x+\eta]\subset(a,b)$, on which $D_1 f$ is uniformly continuous. For $0<|h|<\eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h}=D_1 f(x+\theta h,t)$ with $0<\theta=\theta(h,t)<1$, so

$$\Big|\frac{\phi(x+h)-\phi(x)}{h}-\int_c^d D_1 f(x,t)\,dt\Big|\le\int_c^d |D_1 f(x+\theta h,t)-D_1 f(x,t)|\,dt\le\epsilon(d-c)$$

once $|h|$ is small. ■

Example 8.1

Differentiating $\int_0^1 t^x\,dt=\frac{1}{x+1}$ n times in $x>-1$ gives $\int_0^1 t^x(\log t)^n dt=(-1)^n n!/(x+1)^{n+1}$. Interchanging the order, $\int_a^b\phi(x)\,dx=\int_c^d\int_a^b f(x,t)\,dx\,dt$ for continuous f: both sides, as functions of the upper limit b, vanish at $b=a$ and have the same derivative $\int_c^d f(b,t)\,dt$ (by (a), (b) and Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d\le\infty)$, consider the improper integral $\phi(x)=\int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, and suppose there is $g:[c,d)\to\mathbb{R}$ with (a) $|f(x,t)|\le g(t)$ for all (x,t) and (b) $\int_c^d g<\infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence follows from the comparison test (Proposition 7.8). Given ϵ, choose $d'<d$ with $\int_{d'}^d g<\epsilon$. Then $|\phi(x)-\phi_{d'}(x)|<\epsilon$ for all x, where $\phi_{d'}(x)=\int_c^{d'} f(x,t)\,dt$ is continuous by Proposition 8.1. So ϕ is a uniform limit of continuous functions (Theorem 6.1). ■

Theorem 8.3 (8.1.3)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0\in(a,b)$. If there is $h:[c,d)\to\mathbb{R}$ with (a) $|D_1 f(x,t)|\le h(t)$ and (b) $\int_c^d h<\infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x\in(a,b)$, and ϕ is differentiable with $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. By the mean value theorem, $|f(x,t)-f(x_0,t)| \le |x-x_0|h(t)$, so $\int_c^d f(x,t)dt$ converges by comparison. For $c < d' < d$ let $\phi_{d'}(x) = \int_c^{d'} f(x,t)dt$; then $\phi'_{d'}(x) = \int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), and these converge uniformly in x as $d' \to d$, because the tails are bounded by $\int_{d'}^d h$. Also $\phi_{d'}(x_0) \to \phi(x_0)$. Applying Theorem 6.4 along any sequence $d'_n \to d$ gives the result. ∎

Example 8.2

(a) $F(x) = \int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x) = -\int_0^\infty te^{-t^2}\sin(xt)\,dt = -\frac{x}{2}F(x)$ (integration by parts), so $F(x) = F(0)e^{-x^2/4} = \frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}$: for $G(s) = \int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s>0)$, Theorem 8.3 gives $G'(s) = -\frac{1}{1+s^2}$, and $G(s) \to 0$ as $s \to \infty$, so $G(s) = \frac{\pi}{2} - \arctan s$; continuity at $s=0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

For $x > 0$ let $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t} \le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x) = \int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$.

A function $f : I \to \mathbb{R}$ is **convex** if $f((1-\lambda)x+\lambda y) \le (1-\lambda)f(x) + \lambda f(y)$ for $x, y \in I$, $0 \le \lambda \le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f : I \to \mathbb{R}$ with $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ for all $x, y \in I$ is convex.
(b) A convex f satisfies, for $s < t < u$ in I,

$$\frac{f(t)-f(s)}{t-s} \le \frac{f(u)-f(s)}{u-s} \le \frac{f(u)-f(t)}{u-t}.$$

Proof. (a) By induction on n, the convexity inequality holds for dyadic $\lambda = k/2^n$: if $\lambda = \frac{\lambda_1+\lambda_2}{2}$ with dyadic λ_i of level $n-1$, then $(1-\lambda)x+\lambda y$ is the midpoint of the points for λ_1, λ_2. For general λ, approximate by dyadic $\lambda_n \to \lambda$ and use continuity.

(b) Write $t = (1-\lambda)s + \lambda u$ with $\lambda = \frac{t-s}{u-s}$; then $f(t) \le (1-\lambda)f(s) + \lambda f(u)$, which rearranges to each of the two inequalities. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big) = \int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt \le \Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f : (0,\infty) \to (0,\infty)$ satisfies (a) $f(x+1) = xf(x)$, (b) $f(1) = 1$, (c) $\log f$ is convex, then $f = \Gamma$.

Proof. Γ has these properties, so it suffices to show f is determined by them; by (a) it suffices to determine f on $(0,1]$. Let $\phi = \log f$, $0 < x \le 1$, $n \in \mathbb{N}$. By (a), (b), $\phi(n+1) = \log n!$. Apply Proposition 8.4(b) to the points $n < n+1 < n+1+x$ and $n+1 < n+1+x < n+2$:

$$\log n \le \frac{\phi(n+1+x)-\phi(n+1)}{x} \le \log(n+1).$$

By (a), $\phi(n+1+x) = \phi(x) + \log[x(x+1)\cdots(x+n)]$. Hence

$$0 \le \phi(x) - \log \frac{n!\, n^x}{x(x+1)\cdots(x+n)} \le x \log\Big(1 + \frac{1}{n}\Big) \to 0,$$

so $f(x) = \lim_n \frac{n!\, n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ■

Example 8.3

The **beta function** $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac{1}{2})^2 = B(\frac{1}{2},\frac{1}{2}) = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \Gamma(\frac{1}{2}) = \sqrt{\pi}$. Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\, x^x e^{-x}$ follows from the substitution $t = x(1+u)$ and Laplace's method.

Chapter 9. Fourier Series

9.1 Fourier coefficients

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n} \hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\|x - \sum_{n=-N}^{N} c_n\phi_n\Big\| \le \Big\|x - \sum_{n=-N}^{N} z_n\phi_n\Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}}|c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x-s, \phi_n\rangle = 0$ for $|n| \le N$, so $x-s \perp s-t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum|c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form, $s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n}(a_k\cos kx + b_k \sin kx)$ with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum(-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}} \frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k \text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty}\int_I f(t)\sin(\alpha t + \beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t+\beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I|f-s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ∎

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the **Dirichlet kernel**:

$$s_n(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t) = \sum_{|k|\le n} e^{ikt} = \frac{\sin(n+\frac{1}{2})t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n = 1.$$

Since D_n is even, $s_n(f;x) = \frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0 < \delta < \pi$ and $\int_{-\pi}^{\pi}|f| < \infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac{1}{2})t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)} - \frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)} - \frac{2}{t})\sin(n+\frac{1}{2})t\,dt \to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac{1}{2})t}{t}$ also tends to 0. ■

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt = \lim\int_0^{\lambda\delta}\frac{\sin u}{u}du = \frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt < \infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Write $g(t) = g(0+) + (g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ■

Theorem 9.6 (9.2.4, Dini's test)

Let $f \in \mathcal{R}^1[-\pi,\pi]$ and $x \in [-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt < \infty$. Then $s_n(f;x) \to \frac{1}{2}[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$, with $\lambda = n+\frac{1}{2}$. ■

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)| \le Mt$ for small $|t|$, then $s_n(f;x) \to f(x)$.

Proof. The integrand in (b) is at most $2M$. ■

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x) \to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)| \le (|f'(x)|+1)|t|$ for small t. ■

Lemma 9.9 (9.2.7)

If $g:[0,\delta] \to \mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. By Jordan decomposition we may assume g increasing, and subtracting a constant, $g(0+) = 0$. Given ϵ, choose $h \in (0, \delta)$ with $g(h) < \epsilon$. On $[h, \delta]$, $g(t)/t$ is integrable, so that part tends to 0 by Theorem 9.3. On $(0, h]$, by the second mean value theorem (Corollary 5.25 of Volume I, applied on $[\eta, h]$ and letting $\eta \to 0$), $\int_0^h g(t)\frac{\sin \lambda t}{t}dt = g(h-)\int_c^h \frac{\sin \lambda t}{t}dt$ for some $c \in [0, h]$. Since $|\int_a^b \frac{\sin u}{u}du| \le K$ for all $0 \le a < b$ (a bounded quantity, as $\int_0^\infty \frac{\sin u}{u}$ converges), this part is at most ϵK. ■

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f \in \mathcal{R}^1[-\pi, \pi]$ is of bounded variation on some interval $[x-\delta, x+\delta]$, then $s_n(f; x) \to \frac{1}{2}[f(x+) + f(x-)]$.

Proof. $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0, \delta]$ with $g(0+) = \frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ■

Example 9.2

For $f(x) = x$ on $(-\pi, \pi)$ the series converges to x inside and to 0 at $\pm\pi$; at $x = \pi/2$ it gives $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon). There exist continuous functions whose Fourier series diverge at a point (du Bois-Reymond).

9.3 Fejér's theorem

The **Cesàro means** are $\sigma_n(f; x) = \frac{1}{n+1}\sum_{k=0}^n s_k(f; x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t) = \frac{1}{n+1}\sum_{k=0}^{n} D_k(t) = \frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n \ge 0$, $\frac{1}{2\pi}\int K_n = 1$, and $K_n(t) \le \frac{1}{(n+1)\sin^2(\delta/2)} \to 0$ uniformly on $\delta \le |t| \le \pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t \cdot \sin\frac{t}{2} = \frac{1}{2}\sum(\cos kt - \cos(k+1)t) = \sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f \in \mathcal{R}^1[-\pi, \pi]$ and $f(x\pm)$ exist, then $\sigma_n(f; x) \to \frac{1}{2}[f(x+) + f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f) \to f$ uniformly.

Proof. Since K_n is even, $\sigma_n(f; x) - s = \frac{1}{2\pi}\int_0^\pi [f(x+t)+f(x-t)-2s]K_n(t)\,dt$ with $s = \frac{1}{2}[f(x+)+f(x-)]$. Given ϵ, choose δ with $|f(x+t)+f(x-t)-2s| < \epsilon$ for $0 < t < \delta$. The integral over $(0, \delta)$ is at most $\epsilon \cdot \frac{1}{2\pi}\int_0^\pi K_n \le \epsilon/2$, and that over $[\delta, \pi]$ is at most $\frac{1}{(n+1)\sin^2(\delta/2)} \cdot \frac{1}{2\pi}\int |f(x+t)+f(x-t)-2s| \to 0$. For continuous periodic f, δ can be chosen uniformly in x and f is bounded, giving uniform convergence. ■

So every continuous periodic function is a uniform limit of trigonometric polynomials (the trigonometric form of Weierstrass's theorem), and a continuous periodic function is determined by its Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \geq N$. For $n > N$, $|\sigma_n| \leq \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Given ϵ, choose a continuous g vanishing near $\pm\pi$ (so continuous as a periodic function) with $\|f - g\|_2 < \epsilon$ (Theorem 7.14 and Lemma 7.13), and by Theorem 9.11 a trigonometric polynomial $T = \sigma_m(g)$ of degree m with $\|g - T\|_{\sup} < \epsilon$, so $\|g - T\|_2 < \epsilon$. For $n \geq m$, Theorem 9.1 gives $\|f - s_n(f)\|_2 \leq \|f - T\|_2 < 2\epsilon$. ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a + 2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I|f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\leq n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k\|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2dx = \frac{\pi^2}{3} = 2\sum_{n\geq 1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum\hat{f}(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat{f}(n)\int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $|\int_a^x(f - s_n(f))| \leq \sqrt{2\pi}\left(\int_I|f - s_n(f)|^2\right)^{1/2} \to 0$ by Corollary 9.14. ■

Note that the integrated series converges whether or not the Fourier series of f converges. If f is periodic and C^1, integration by parts gives $\widehat{f'}(n) = in\hat{f}(n)$; since $\sum|n\hat{f}(n)|^2 < \infty$, Cauchy–Schwarz gives $\sum|\hat{f}(n)| < \infty$, so the Fourier series converges uniformly (to f, by Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat{f}(n) = a(n)$.

Proof. Repeated integration by parts gives $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$, so $|n^k \hat{f}(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely the series and all its term-by-term derivatives $\sum (ik)^j a(k) e^{ikx}$ converge uniformly by the M-test (as $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4); integrating the uniformly convergent series against e^{-inx} term by term gives $\hat{f}(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a,b) \to \mathbb{R}$ is analytic and $K \subset (a,b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \dots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k (x-c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k \ge n} \frac{k!}{(k-n)!} |a_k| \Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n} n! \sum_{j \ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n! \Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \dots$, then $c \le (R+1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f : \mathbb{R} \to \mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty} |\hat{f}(n)|^{1/|n|} < 1$.

Proof. By Lemma 9.18 on $K = [-\pi, \pi]$, $\|f^{(k)}\|_{\sup} \le Ck!R^k$ for $k \ge 1$ (and enlarging C, also for $k = 0$). Then $|\hat{f}(n)| = |\widehat{f^{(k)}}(n)|/|n|^k \le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat{f}(n)| \le C(R+1)e^{-|n|/(R+1)}$, and $\limsup |\hat{f}(n)|^{1/|n|} \le e^{-1/(R+1)} < 1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi} |f| < \infty$, then $\lim_{|n|\to\infty} \hat{f}(n) = 0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i \sin nt)\, dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f \in \mathcal{R}^1[-\pi, \pi]$ is odd, then the partial sums of $\sum_{n\geq 1} \hat{f}(n)/n$ are bounded.

Proof. For odd f, $\hat{f}(n) = -\frac{i}{\pi}\int_0^\pi f(t)\sin nt\,dt$, so $\sum_{n\leq N}\frac{\hat{f}(n)}{n} = -\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\leq N}\frac{\sin nt}{n}\,dt$. The sums $\sum_{n\leq N}\frac{\sin nt}{n}$ are bounded by $1+2\pi$ on $(0,\pi]$: the terms with $n \leq 1/t$ are at most t each, and for $n > 1/t$, summation by parts with (‡) of Chapter 6 gives at most $2\cdot t\cdot \pi/t$. Hence the partial sums are bounded by $\frac{1+2\pi}{\pi}\int_0^\pi |f|$. ■

So $\sum_{n\geq 2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f \in \mathcal{R}^1$, since $\sum \frac{1}{n\log n} = \infty$.

Chapter 10. The Lebesgue Integral

10.1 Measurable sets and Lebesgue measure

Proposition 10.1 (10.1.1)

Every open $U \subset \mathbb{R}$ is a union of countably many disjoint open intervals, and this representation is unique.

Proof. For $x \in U$ let $I_x = (a_x, b_x)$ with $a_x = \inf\{a : (a, x] \subset U\}$, $b_x = \sup\{b : [x, b) \subset U\}$; $I_x \subset U$ is the largest open interval in U containing x. If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval in U, so $I_x = I_y$. Each I_x contains a rational, and distinct ones are disjoint, so there are countably many. The intervals are the components of U, which gives uniqueness. ■

For $U = \bigsqcup(a_n, b_n)$ put $\lambda(U) = \sum(b_n - a_n)$ (independent of the order, Proposition 2.16 of Volume I), and for any $S \subset \mathbb{R}$ define the **outer measure**

$$\mu(S) = \inf\{\lambda(U) : U \supset S \text{ open}\} \in [0, \infty].$$

Proposition 10.2 (10.1.2)

(a) $S \subset T \Rightarrow \mu(S) \leq \mu(T)$.
(b) If I is an interval with end points $a \leq b$, then $\mu(I) = b - a$.
(c) $\mu(x + S) = \mu(S)$.
(d) $\mu(\bigcup S_n) \leq \sum \mu(S_n)$ for countably many S_n.

Proof. First, if an open interval (a, b) is covered by open intervals J_k, then $b - a \leq \sum |J_k|$: for $a < c < d < b$ finitely many J_k cover the compact $[c, d]$, and by choosing a chain of overlapping intervals from c to d, $d - c \leq \sum |J_k|$; let $c \to a$, $d \to b$. Summing over components (Proposition 6.9), $\lambda(U) \leq \sum \lambda(U_n)$ whenever $U \subset \bigcup U_n$ are open.

(a) is immediate.

(b) For $I = [a, b]$, $\mu(I) \leq \lambda((a - \epsilon, b + \epsilon)) = b - a + 2\epsilon$; and an open $U \supset I$ has a component containing $[a, b]$, so $\lambda(U) \geq b - a$. Other intervals are similar.

(c) Translation preserves λ of open sets.

(d) We may assume all $\mu(S_n) < \infty$. Choose open $U_n \supset S_n$ with $\lambda(U_n) < \mu(S_n) + \epsilon 2^{-n}$; then $\mu(\bigcup S_n) \leq \lambda(\bigcup U_n) \leq \sum \lambda(U_n) < \sum \mu(S_n) + \epsilon$. ■

In particular countable sets have outer measure 0, so $[0, 1]$ is uncountable. A set $E \subset \mathbb{R}$ is **measurable** if

$$\mu(X) = \mu(X \cap E) + \mu(X \cap E^c) \qquad \text{for every } X \subset \mathbb{R};$$

let $\mathfrak{M}$ be the family of measurable sets. By (d), only $\geq$ needs checking, and only for $\mu(X) < \infty$.

Theorem 10.3 (10.1.3)

(a) $\emptyset \in \mathfrak{M}$.
(b) $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$.
(c) $E, F \in \mathfrak{M} \Rightarrow E \cup F \in \mathfrak{M}$.
(d) $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$.
(e) If $\mu(E) = 0$, then $E \in \mathfrak{M}$.
(f) Every interval is measurable.

(g) Every open set is measurable.

Proof. (a), (b) are clear from the symmetric definition.

(c) Since $X \cap (E \cup F) = (X \cap E) \cup (X \cap F \cap E^c)$,

$$\mu(X\cap(E\cup F))+\mu(X\cap(E\cup F)^c) \le \mu(X\cap E)+\mu(X\cap E^c\cap F)+\mu(X\cap E^c\cap F^c) = \mu(X\cap E)+\mu(X\cap E^c) = \mu(X),$$

using $F \in \mathfrak{M}$ (for the set $X \cap E^c$) and then $E \in \mathfrak{M}$.

(d) Let $F_n = E_n - (E_1 \cup \cdots \cup E_{n-1})$ (measurable by (b), (c), as $A \cap B = (A^c \cup B^c)^c$), disjoint, and $G_n = F_1 \cup \cdots \cup F_n$, $E = \bigcup E_n = \bigsqcup F_n$. Using $F_n \in \mathfrak{M}$ for the set $X \cap G_n$, $\mu(X \cap G_n) = \mu(X \cap G_{n-1}) + \mu(X \cap F_n)$, so $\mu(X \cap G_n) = \sum_{i\le n} \mu(X \cap F_i)$. As $G_n \in \mathfrak{M}$ and $G_n^c \supset E^c$,

$$\mu(X) = \mu(X \cap G_n) + \mu(X \cap G_n^c) \ge \sum_{i\le n} \mu(X \cap F_i) + \mu(X \cap E^c).$$

Let $n \to \infty$ and use (d) of Proposition 10.2: $\mu(X) \ge \mu(X \cap E) + \mu(X \cap E^c)$.

(e) $\mu(X \cap E) = 0$ and $\mu(X \cap E^c) \le \mu(X)$.

(f) By (b), (d) and (e) it suffices to treat $I = (a, \infty)$. Given X and ϵ, take open $U \supset X$ with $\lambda(U) < \mu(X)+\epsilon$. The open sets $U\cap(a,\infty)$, $U\cap(-\infty,a)$ satisfy $\lambda(U\cap(a,\infty))+\lambda(U\cap(-\infty,a)) \le \lambda(U)$ (each component of U is split into two intervals), so $\mu(X \cap I) + \mu(X \cap I^c) \le \lambda(U \cap (a,\infty)) + \lambda(U \cap (-\infty, a)) + \mu(\{a\}) < \mu(X) + \epsilon$.

(g) Proposition 10.1, (f) and (d). ■

So $\mathfrak{M}$ is a σ-algebra containing all Borel sets. The **Lebesgue measure** is $m = \mu|_{\mathfrak{M}}$.

Theorem 10.4 (10.1.4, countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigsqcup E_n) = \sum m(E_n)$.

Proof. In the proof of Theorem 10.3(d) take $X = E = \bigsqcup E_n$ (so $F_n = E_n$): $m(E) \ge \sum m(E_n)$; the reverse is Proposition 10.2(d). ■

Consequences: if $E_1 \subset E_2 \subset \cdots$, then $m(\bigcup E_n) = \lim m(E_n)$ (write the union as $\bigsqcup(E_n - E_{n-1})$); if $E_1 \supset E_2 \supset \cdots$ and $m(E_1) < \infty$, then $m(\bigcap E_n) = \lim m(E_n)$. By definition of μ, every measurable E satisfies $m(E) = \inf\{m(U) : U \supset E \text{ open}\}$, and for bounded E, $m(E) = \sup\{m(K) : K \subset E \text{ compact}\}$ (apply the outer approximation to $[a,b] - E$). Not every set is measurable (Vitali): choosing one point in $[0,1)$ from each coset of $\mathbb{Q}$ gives a set V whose rational translates mod 1 are disjoint and cover $[0,1)$, so $m(V)$ could be neither 0 nor positive.

10.2 Measurable functions and the integral

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$. $f : E \to \mathbb{R}^*$ is **measurable** if $\{x \in E : f(x) > a\} \in \mathfrak{M}$ for every $a \in \mathbb{R}$; a complex function is measurable if its real and imaginary parts are.

Proposition 10.5 (10.2.1)

For $f : E \to \mathbb{R}^*$ the following are equivalent:
(a) f is measurable;
(b) $\{f \ge a\} \in \mathfrak{M}$ for all a;
(c) $\{f < a\} \in \mathfrak{M}$ for all a;
(d) $\{f \le a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \ge a\} = \bigcap_n \{f > a - 1/n\}$, $\{f > a\} = \bigcup_n \{f \ge a + 1/n\}$, and (c), (d) are the complements (relative to E) of (b), (a). ∎

Corollary 10.6 (10.2.2)

If f is measurable, then $\{f = a\}$ is measurable for every $a \in \mathbb{R}^*$.

Proof. $\{f = a\} = \{f \ge a\} \cap \{f \le a\}$ for real a, and $\{f = \infty\} = \bigcap_n \{f \ge n\}$. ∎

Proposition 10.7 (10.2.3)

(a) Continuous $f : E \to \mathbb{R}$ are measurable.
(b) If f, g are measurable and real-valued and $a \in \mathbb{R}$, then $f + g$, fg, af are measurable.
(c) If each f_n is measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf f_n$, $\limsup f_n$, and hence $\lim f_n$ where it exists.

Proof. (a) $\{f > a\} = U \cap E$ with U open.

(b) $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > r\} \cap \{g > a - r\}$; f^2 is measurable since $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\}$ for $a \ge 0$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

(c) $\{\sup f_n > a\} = \bigcup \{f_n > a\}$, $\inf f_n = -\sup(-f_n)$, $\limsup f_n = \inf_n \sup_{k \ge n} f_k$. ∎

A **simple function** is $s = \sum_{i=1}^k a_i \chi_{A_i}$ with $a_i \ge 0$ and $A_i \in \mathfrak{M}$; its integral is $\int_E s = \sum a_i m(A_i \cap E)$ (independent of the representation, by passing to a common disjoint refinement and using additivity). For measurable $f : E \to [0, \infty]$,

$$\int_E f = \sup\Big\{ \int_E s : 0 \le s \le f,\ s \text{ simple} \Big\}.$$

For simple functions the two definitions agree, and $0 \le f \le g$ implies $\int f \le \int g$.

Proposition 10.8 (10.2.4)

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions.

Proof. Let $A_{n,i} = \{\frac{i-1}{2^n} \le f < \frac{i}{2^n}\}$ $(1 \le i \le n2^n)$, $B_n = \{f \ge n\}$, and $s_n = \sum_i \frac{i-1}{2^n} \chi_{A_{n,i}} + n\chi_{B_n}$. Then $s_n \le s_{n+1} \le f$, and $f - s_n < 2^{-n}$ where $f < n$; also $s_n = n \to \infty$ where $f = \infty$. ∎

Theorem 10.9 (10.2.5)

If $A = \bigsqcup A_n$ with $A_n \in \mathfrak{M}$ disjoint and $f : A \to [0, \infty]$ is measurable, then $\int_A f = \sum_n \int_{A_n} f$.

Proof. For simple s, $\int_A s = \sum_n \int_{A_n} s$ by Theorem 10.4; so for simple $s \le f$, $\int_A s \le \sum \int_{A_n} f$, and taking the supremum, $\int_A f \le \sum \int_{A_n} f$. Conversely, given N and ϵ, choose simple $s_k \le f$ on A_k with $\int_{A_k} s_k > \int_{A_k} f - \epsilon/N$ $(k \le N)$; $s = \sum_{k \le N} s_k \chi_{A_k} \le f$, so $\int_A f \ge \sum_{k \le N} \int_{A_k} f - \epsilon$. ∎

So $A \mapsto \int_A f$ is a measure. For measurable f, write $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$; $f \in L^1(E)$ if $\int_E |f| < \infty$, and then $\int_E f = \int_E f^+ - \int_E f^-$; for complex f, $\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f$.

Corollary 10.10 (10.2.6)

Theorem 10.9 holds for every $f \in L^1(A)$, the series converging absolutely.

Proof. Apply Theorem 10.9 to $f^{\pm}$ (real and imaginary parts) and subtract; the series converge because $\sum \int_{A_n} |f| = \int_A |f| < \infty$. ■

Corollary 10.11 (10.2.7)

A measurable f is in $L^1(E)$ iff $|f| \in L^1(E)$, and then $|\int_E f| \le \int_E |f|$.

Proof. The first statement is the definition. Choose θ with $e^{-i\theta} \int f = |\int f|$; then $|\int f| = \int \mathrm{Re}(e^{-i\theta} f) \le \int |f|$ (using linearity, Theorem 10.14). ■

Proposition 10.12 (10.2.8)

If $f \ge 0$ is measurable and $\int_E f = 0$, then $f = 0$ almost everywhere.

Proof. With $A_n = \{f \ge 1/n\}$, $\frac{1}{n} m(A_n) \le \int f = 0$, so $m(\{f > 0\}) = m(\bigcup A_n) = 0$. ■

10.3 Convergence theorems

Theorem 10.13 (10.3.1, monotone convergence)

If $0 \le f_1 \le f_2 \le \cdots$ are measurable on E and $f = \lim f_n$, then $\int_E f = \lim \int_E f_n$.

Proof. f is measurable and $\int f_n \le \int f_{n+1} \le \int f$, so $L = \lim \int f_n \le \int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$; $E_n = \{f_n \ge cs\}$ increase to E (where $f(x) = 0$, $x \in E_1$; otherwise $cs(x) < f(x)$). Then $\int f_n \ge \int_{E_n} f_n \ge c \int_{E_n} s \to c \int_E s$ (Theorem 10.9, continuity from below). So $L \ge c \int s$; let $c \to 1$ and take the supremum over s. ■

Theorem 10.14 (10.3.2, linearity)

(a) If $f, g \in L^1(E)$, then $f + g \in L^1(E)$; if $f, g \ge 0$ or $f, g \in L^1(E)$, then $\int (f + g) = \int f + \int g$.
(b) If $f \in L^1(E)$ and $a \in \mathbb{C}$, then $af \in L^1(E)$ and $\int af = a \int f$.

Proof. For simple functions this is immediate. For $f, g \ge 0$ take increasing simple $s_n \to f$, $t_n \to g$ (Proposition 10.8); $s_n + t_n \uparrow f + g$, and Theorem 10.13 gives the formula. Then $|f + g| \le |f| + |g|$ gives integrability. For real $f, g \in L^1$ and $h = f + g$, $h^+ + f^- + g^- = h^- + f^+ + g^+$; integrate and rearrange. Complex functions: real and imaginary parts.

(b) is similar (for $a = i$ the parts are exchanged). ■

Corollary 10.15 (10.3.3)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \sum_n f_n = \sum_n \int_E f_n$.

Proof. Apply Theorem 10.13 to the partial sums, using Theorem 10.14. ■

Theorem 10.16 (10.3.4, Riemann and Lebesgue integrals)

Let $f : [a, b] \to \mathbb{R}$ be bounded.
(a) If f is Riemann integrable, then f is measurable and its Lebesgue integral equals its Riemann integral.

(b) f is Riemann integrable iff the set of its discontinuities has measure 0.

Proof. Choose partitions $P_1 \subset P_2 \subset \cdots$ with $\|P_k\| \to 0$, and let ℓ_k, u_k be the step functions equal to m_i, M_i on the open subintervals of P_k (and to f at partition points). Then $\ell_k \uparrow \ell$, $u_k \downarrow u$, $\ell \le f \le u$, and by monotone convergence (applied to $u_1 - u_k$ and $\ell_k - \ell_1$), $\int \ell = \lim L(f, P_k)$ and $\int u = \lim U(f, P_k)$. Let $P = \bigcup P_k$, a countable set. For $x \notin P$, $u(x)$ and $\ell(x)$ are the limits of sup and inf of f over shrinking neighbourhoods of x, so f is continuous at x iff $u(x) = \ell(x)$.

If f is Riemann integrable, then $U(f, P_k) - L(f, P_k) \to 0$ (proof of Theorem 5.8 of Volume I), so $\int(u - \ell) = 0$ and $u = \ell$ a.e. (Proposition 10.12). Then $f = u$ a.e.; f is measurable (Theorem 10.3(e)) and $\int f = \int u = \lim U(f, P_k)$ is the Riemann integral. This proves (a) and, with the characterization of continuity, the necessity in (b). Conversely, if the discontinuities have measure 0, then $u = \ell$ a.e., so $\lim(U(f, P_k) - L(f, P_k)) = \int(u - \ell) = 0$, and f is Riemann integrable by Riemann's criterion. ■

Theorem 10.17 (10.3.5, Fatou's lemma)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \liminf f_n \le \liminf \int_E f_n$.

Proof. $g_n = \inf_{k \ge n} f_k$ increases to $\liminf f_n$ and $\int g_n \le \int f_k$ for $k \ge n$; apply Theorem 10.13. ■

Theorem 10.18 (10.3.6, Lebesgue's dominated convergence theorem)

Let $f_n : E \to \mathbb{C}$ be measurable with (a) $|f_n| \le g$ for some $g \in L^1(E)$ and (b) $f(x) = \lim f_n(x)$ for each x. Then $f \in L^1(E)$ and $\lim \int_E |f_n - f| = 0$; in particular $\int f_n \to \int f$.

Proof. $|f| \le g$, so $f \in L^1$. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int(2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Hence $\limsup \int |f_n - f| \le 0$. ■

The theorems remain valid when the hypotheses hold only almost everywhere, since sets of measure 0 do not affect integrals.

Theorem 10.19 (10.3.7, Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int_E |f_n| < \infty$, then $\sum f_n(x)$ converges absolutely for almost every x, its sum f is in $L^1(E)$, and $\int_E f = \sum \int_E f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ (Corollary 10.15), so $g < \infty$ a.e. There the series converges absolutely, with partial sums dominated by g; apply Theorem 10.18. ■

Theorem 10.20 (10.3.8, absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ for all measurable $A \subset E$ with $m(A) < \delta$.

Proof. $\min(|f|, n) \uparrow |f|$, so by monotone convergence choose N with $\int(|f| - \min(|f|, N)) < \epsilon/2$, and $\delta = \epsilon/(2N)$. Then $\int_A |f| \le \epsilon/2 + Nm(A) < \epsilon$. ■

10.4 The spaces L^1 and L^2

Identifying functions that agree a.e., $L^1(E)$ with $\|f\|_1 = \int |f|$ and $L^2(E)$ (measurable f with $\int |f|^2 < \infty$) with $\|f\|_2 = (\int |f|^2)^{1/2}$ are normed spaces; L^2 has the inner product $\langle f, g \rangle = \int f\bar{g}$ ($|f\bar{g}| \le \frac{1}{2}(|f|^2 + |g|^2)$).

Theorem 10.21 (10.4.1)

Every Cauchy sequence in $L^1(E)$ converges.

Proof. By Proposition 7.11 it suffices that $\sum\|f_n\|_1 < \infty$ implies convergence of $\sum f_n$ in L^1. By Theorem 10.19, $f = \sum f_n$ exists a.e. and is in L^1, and $\|f - \sum_{n\le N} f_n\|_1 \le \sum_{n>N}\|f_n\|_1 \to 0$. ■

Theorem 10.22 (10.4.2)

$L^2(E)$ is a Banach space (a Hilbert space).

Proof. Let $\sum\|f_n\|_2 = M < \infty$ and $g = \sum|f_n|$. By the triangle inequality $\|\sum_{n\le N}|f_n|\|_2 \le M$, and monotone convergence gives $\int g^2 \le M^2$, so $g < \infty$ a.e. and $f = \sum f_n$ exists a.e. with $|f| \le g$. Since $|f - \sum_{n\le N} f_n|^2 \le g^2$ and tends to 0 a.e., dominated convergence gives $\|f - \sum_{n\le N} f_n\|_2 \to 0$. Apply Proposition 7.11. ■

Theorem 10.23 (10.4.3, Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is a periodic $f \in L^2[-\pi,\pi]$ with $\hat f(n) = c(n)$ for all n.

Proof. By orthonormality, $s_N = \sum_{|n|\le N} c(n)e^{inx}$ satisfies $\frac{1}{2\pi}\int|s_M - s_N|^2 = \sum_{N<|n|\le M}|c(n)|^2$, so (s_N) is Cauchy in L^2 and converges to some f (Theorem 10.22). By Cauchy–Schwarz, $\hat f(n) = \lim_N \widehat{s_N}(n) = c(n)$. ■

Thus $f \mapsto \hat f$ is an isometric isomorphism of $L^2[-\pi,\pi]$ (with $\frac{1}{2\pi}\int$) onto $\ell^2(\mathbb{Z})$, once injectivity is known (Theorem 10.26).

Proposition 10.24 (10.4.4)

If $f \in L^1(E)$ and $\int_A f = 0$ for every measurable $A \subset E$, then $f = 0$ a.e.

Proof. With $A = \{\operatorname{Re} f > 0\}$, $\int_A \operatorname{Re} f = \operatorname{Re}\int_A f = 0$, so $\operatorname{Re} f\chi_A = 0$ a.e. (Proposition 10.12) and $m(A) = 0$. Similarly for $\{\operatorname{Re} f < 0\}$, $\{\operatorname{Im} f > 0\}$, $\{\operatorname{Im} f < 0\}$. ■

Lemma 10.25 (10.4.5)

For every measurable $E \subset [a,b]$ there are continuous g_n with $|g_n| \le 1$ and $g_n \to \chi_E$ a.e.

Proof. By regularity choose compact K_n and open U_n with $K_n \subset E \subset U_n$ and $m(U_n - K_n) < 2^{-n}$, and let $g_n(x) = \frac{d(x,U_n^c)}{d(x,U_n^c)+d(x,K_n)}$, continuous, equal to 1 on K_n and 0 off U_n. Then $g_n(x) \ne \chi_E(x)$ only for $x \in U_n - K_n$. The set of x lying in infinitely many $U_n - K_n$ is contained in $\bigcup_{n\ge N}(U_n - K_n)$ for every N, so has measure at most 2^{1-N}, i.e. 0. Elsewhere $g_n(x) \to \chi_E(x)$. ■

Theorem 10.26 (10.4.6)

If $f \in L^1[-\pi,\pi]$ and $\hat f(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.

Proof. By hypothesis $\int fg = 0$ for every trigonometric polynomial g, hence for every continuous periodic g (Fejér, Theorem 9.11, gives uniformly convergent trigonometric polynomials; pass to the limit by dominated convergence). Let $A \subset [-\pi+\eta, \pi-\eta]$ be measurable. Lemma 10.25, with the U_n inside $(-\pi,\pi)$, gives continuous periodic g_n, $|g_n| \le 1$, $g_n \to \chi_A$ a.e.; by dominated convergence

$\int_A f = \lim \int f g_n = 0$. Letting $\eta \to 0$, $\int_A f = 0$ for all measurable A, and Proposition 10.24 applies. ■

Hence two L^1 functions with the same Fourier coefficients agree a.e., and Parseval's identity holds for every $f \in L^2[-\pi, \pi]$.

Lemma 10.27 (10.4.7)

For $f \in L^2(E)$ and $\epsilon > 0$ there is a simple function s (a finite complex combination of characteristic functions of sets of finite measure) with $\|f - s\|_2 < \epsilon$.

Proof. For $f \geq 0$ take increasing simple $s_n \to f$ (Proposition 10.8); $|f - s_n|^2 \leq f^2$, so $\|f - s_n\|_2 \to 0$ by dominated convergence, and $\int s_n^2 < \infty$ forces the sets in s_n to have finite measure. In general use $f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-$. ■

Theorem 10.28 (10.4.8)

For $f \in L^2[a, b]$ and $\epsilon > 0$ there is $g \in C[a, b]$ with $\|f - g\|_2 < \epsilon$.

Proof. By Lemma 10.27 it suffices to approximate χ_A, $A \subset [a, b]$ measurable. With g_n from Lemma 10.25, $|g_n - \chi_A|^2 \leq 4$ and tends to 0 a.e., so $\|g_n - \chi_A\|_2 \to 0$ by dominated convergence on the finite interval. ■

Theorem 10.29 (10.4.9)

For $f \in L^1(E)$ and $\epsilon > 0$ there is a continuous g on $\mathbb{R}$ vanishing outside a bounded interval with $\|f - g\|_1 < \epsilon$ (norm taken over E).

Proof. By dominated convergence $\|f - f\chi_{[-N,N]}\|_1 \to 0$; approximate $f\chi_{[-N,N]}$ in L^1 by simple functions as in Lemma 10.27; approximate each $\chi_{A \cap E}$, $A \subset [-N, N]$, by g_n from Lemma 10.25 (with $U_n \subset (-N - 1, N + 1)$), using dominated convergence. ■

In particular L^1 and L^2 are the completions of the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ of Chapter 7.

10.5 The Poisson summation formula

For $g \in L^1(\mathbb{R})$ let $G(x) = \sum_{n \in \mathbb{Z}} g(x + 2\pi n)$.

Theorem 10.30 (10.5.1)

The series defining G converges absolutely for almost every x, $G \in L^1[-\pi, \pi]$, and

$$\hat{G}(n) = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(x) e^{-inx}\, dx, \qquad n \in \mathbb{Z}.$$

Proof. Integrals over $\mathbb{R}$ are translation invariant (Proposition 10.2(c) for sets, hence for simple functions and by monotone convergence for all $f \geq 0$). By Corollary 10.15, $\int_{-\pi}^{\pi} \sum_n |g(x + 2\pi n)|\, dx = \sum_n \int_{-\pi + 2\pi n}^{\pi + 2\pi n} |g| = \int_{\mathbb{R}} |g| < \infty$. So the series converges absolutely a.e. and $G \in L^1$. By Theorem 10.19,

$$\hat{G}(n) = \frac{1}{2\pi} \sum_k \int_{-\pi}^{\pi} g(x + 2\pi k) e^{-inx} dx = \frac{1}{2\pi} \sum_k \int_{-\pi + 2\pi k}^{\pi + 2\pi k} g(y) e^{-iny} dy = \frac{1}{2\pi} \int_{\mathbb{R}} g e^{-iny} dy.$$

■

If moreover the Fourier series of G converges to $G(0)$ at 0 (for example if g is smooth and rapidly decreasing), this gives the **Poisson summation formula** $\sum_n g(2\pi n) = \frac{1}{2\pi}\sum_n \int_{\mathbb{R}} g(x)e^{-inx}dx$. Applied to $g(x) = e^{-ax^2}$ it yields the transformation formula for the theta function $\theta(x) = \sum_n e^{-\pi n^2 x}$: $\theta(1/x) = \sqrt{x}\,\theta(x)$.